Md Ershadul Haque

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SUMMARY

Computational engineering researcher specializing in discontinuous Galerkin (DG) finite element methods for radiative transfer and electromagnetic scattering in complex and heterogeneous media. Experienced in developing high-fidelity solvers with multilevel adaptive mesh refinement in both space and angle, multigrid acceleration, and scalable parallel implementation on HPC systems using deal.II, MPI, and p4est. Develops graph neural network (GNN) surrogates for scattering and phase matrix computation that couple directly into DG radiative transfer solvers. Additional background in radiation and convection heat transfer, numerical solution of PDEs, and industrial process engineering.

EDUCATION

The University of Tennessee, Knoxville (UTK) Knoxville, TN, USA
Ph.D. in Mechanical & Aerospace Engineering *Jun 2022 - Jan 2027*
GPA: 3.91 / 4.00

Dissertation Title: *The Use of GNN for Scattering and Phase Matrices Calculations and Their Incorporation in Discontinuous Galerkin Solution of the Radiative Transfer Equation*

King Fahd University of Petroleum and Minerals (KFUPM) Dhahran, Saudi Arabia
M.S. in Mechanical Engineering *Jan 2019 - Dec 2021*
GPA: 3.656 / 4.00

Thesis Title: *The Solution of Radiative Transfer Equation in 2D Complex Geometries Using Body-Fitted Coordinates*

Rajshahi University of Engineering & Technology (RUET) Rajshahi, Bangladesh
B.S. in Mechanical Engineering *April 2010 - Dec 2014*
GPA: 3.62 / 4.00

RESEARCH EXPERIENCE AND ACADEMIC APPOINTMENTS

The University of Tennessee, Knoxville (UTK) Knoxville, TN, USA
Graduate Research Assistant *Jun 2022 - Present*
Advisor: *Dr. Reza Abedi*

- Develop space-angle discontinuous Galerkin finite element solvers in C++ (deal.II) for the radiative transfer equation in heterogeneous and semi-transparent media
- Implement multilevel adaptive mesh refinement in both spatial and angular domains to resolve sharp radiative intensity features at reduced computational cost
- Apply multigrid techniques to accelerate convergence of large-scale space-angle discretizations
- Implement parallel solvers using MPI and p4est with combined angular and domain decomposition, enabling scalable execution on HPC clusters
- Model semitransparent boundary conditions and graded-index (continuously variable refractive index) media in 1D plane-parallel and 2D geometries
- Develop graph neural network (GNN) surrogates in PyTorch Geometric for scattering and phase matrix computation and integrate them into the DG radiative transfer solver

The University of Tennessee, Knoxville (UTK) Knoxville, TN, USA
Graduate Teaching Assistant *Jan 2025 - May 2025*

- Teaching Assistant for ME 317: Heat Transfer, ME 321: mechanics of Materials, AE 341/347: Fluid Mechanics I; held help sessions for students, prepared quiz questions, and graded midterm and final exams

King Fahd University of Petroleum and Minerals (KFUPM) Dhahran, Saudi Arabia
Teaching Assistant, Mechanical Engineering Department *Sep 2021 - Dec 2021*

- Instructor for ME 217: Materials Lab

King Fahd University of Petroleum and Minerals (KFUPM) Dhahran, Saudi Arabia
Research Assistant, City and Regional Planning Department *Sep 2020 - Dec 2021*

- Solved systems of partial differential equations (PDEs) numerically in MATLAB for kinetic and reactor modeling of naphtha catalytic reforming
- Performed literature review and model parameter estimation for the reforming kinetic model

King Fahd University of Petroleum and Minerals (KFUPM) Dhahran, Saudi Arabia
Grader, Mechanical Engineering Department *Sep 2019 - May 2021*

- Graded homework for mechanical engineering courses and served as exam proctor

INDUSTRY EXPERIENCE

Walton Hi-Tech Industries Ltd., Compressor R&D Dhaka, Bangladesh
Process Engineer *May 2017 - Dec 2018*

- Handled product implementation and process engineering for compressor manufacturing
- Supported automated lathe, milling, drilling, and grinding lines for manufacturing compressor pistons and connecting rods
- Supported automatic resistance welding and washing/phosphating lines

Mason Acoustic Company Ltd., Sound Treatment & Fireproof Door Dhaka, Bangladesh
Sales and Service Engineer *Aug 2016 - Apr 2017*

- Worked toward sales targets for acoustic treatment and fireproof door products
- Inspected, installed, and maintained acoustic treatment in generator rooms, auditoriums, and TV studios

Ashuganj Power Station Company Ltd. Brahmanbaria, Bangladesh
Industrial Intern

- Studied operation and maintenance across plant workstations, including steam boilers, steam and gas turbines, gas engines, fans, compressors, condensers, and pumps

RESEARCH INTERESTS

- Discontinuous Galerkin Finite Element Methods
- Electromagnetic Scattering
- High-Performance Parallel Computing
- Numerical Analysis of PDEs
- Radiative Transfer and Radiation Heat Transfer
- Adaptive Mesh Refinement and Multigrid Methods
- Scientific Machine Learning (GNNs)
- Nanofluids and Renewable Energy Systems

TECHNICAL SKILLS

Programming Languages: C++, Python, MATLAB
FEM Frameworks & Libraries: deal.II, p4est
Simulation Software (Commercial): ANSYS, COMSOL, EES
Parallel Computing & Tools: MPI, HPC clusters, Git
Machine Learning: PyTorch, PyTorch Geometric, graph neural networks (GNNs)
Statistics: MINITAB, regression analysis, design of experiments
Modeling & CAD: SolidWorks, AutoCAD

PUBLICATIONS AND PRESENTATIONS

Journal

1. Wang, H., **Haque, M. E.**, Abedi, R., and Mudaliar, S. *Integrating Angular and Domain Decomposition with Space-Angle Discontinuous Galerkin Methods in 2D Radiative Transfer.* Journal of Quantitative Spectroscopy and Radiative Transfer 329 (2024): 109208.
2. **Haque, M. E.**, Mansoor, S. B., and Yilbas, B. S. *Computational Radiative Transport in Complex Geometries Using Orthogonal Coordinates.* Journal of Non-Equilibrium Thermodynamics (2023).
3. Shafiullah, M., Refat, A. M., **Haque, M. E.**, Chowdhury, D. M. H., Hossain, M. S., Alharbi, A. G., Alam, M. S., Ali, A., and Hossain, S. *Review of Recent Developments in Microgrid Energy Management Strategies.* Sustainability 14 (2022): 14794.

4. Rahman, S. M., Al-Ismail, F. S. M., **Haque, M. E.**, Shafiullah, M., Islam, M. R., Chowdhury, M. T., Alam, M. S., Razzak, S. A., Ali, A., and Khan, Z. A. *Electricity Generation in Saudi Arabia: Tracing Opportunities and Challenges to Reducing Greenhouse Gas Emissions.* IEEE Access 9 (2021): 116163-116182.
5. **Haque, M. E.**, Hossain, M. S., and Ali, H. M. *Laminar Forced Convection Heat Transfer of Nanofluids Inside Non-Circular Ducts: A Review.* Powder Technology (2020).

Book Chapter

1. Shafiullah, M., **Haque, M. E.**, Hossain, S., Hossain, M. S., and Rana, M. J. *Community Microgrid Energy Scheduling Based on the Grey Wolf Optimization Algorithm.* In Artificial Intelligence-Based Energy Management Systems for Smart Microgrids, pp. 47-73. CRC Press, 2022.

Conference

1. **Haque, M. E.**, Abedi, R., and Mudaliar, S. *2D Graded-Index Effects in Radiative Transfer: A Discontinuous Galerkin Method.* In USNC-URSI National Radio Science Meeting (NRSM), pp. 327-328. IEEE, 2026.
2. **Haque, M. E.**, Wang, H., Abedi, R., and Mudaliar, S. *Solution of Radiative Transfer Equation for Plane-Parallel Medium Using a Discontinuous Galerkin Method and Adaptive Mesh Refinement.* In USNC-URSI National Radio Science Meeting (NRSM). IEEE, 2025.
3. **Haque, M. E.**, Wang, H., Abedi, R., and Mudaliar, S. *Discontinuous Galerkin Method for Radiation Transfer in Plane-Parallel Semi-Transparent Media with Continuous Variable Refractive Index.* In USNC-URSI National Radio Science Meeting (NRSM), pp. 110-111. IEEE, 2024.
4. **Haque, M. E.**, Wang, H., Abedi, R., and Mudaliar, S. *A Parallel Implementation of Solving Radiative Transfer Equation Using Discontinuous Galerkin Method.* In IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting, Portland, OR, USA, pp. 1811-1812, 2023.
5. **Haque, M. E.**, Wang, H., Abedi, R., and Mudaliar, S. *A Discontinuous Galerkin Method for Solving Radiative Transfer Equation with Semitransparent Boundary Conditions.* In USNC-URSI National Radio Science Meeting (NRSM), Boulder, CO, USA, pp. 42-43, 2023.
6. Shafiullah, M., Rana, M. J., **Haque, M. E.**, Islam, A., Rahman, S. M., Alam, M. S., and Ali, A. *An Intelligent Approach for Power Quality Events Detection and Classification.* In 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA), IEEE, pp. 194-199, 2021. (Best Paper Award)
7. Shafiullah, M., **Haque, M. E.**, Al-Ismail, F. S., Islam, A., Alam, M. S., Ali, A., and Rahman, S. M. *Backtracking Search Algorithm for PV Module Electrical Parameter Estimation.* In 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA), IEEE, pp. 188-193, 2021.

AWARDS AND HONORS

Graduate Student Senate Travel Award, UTK (2023-2026)
Travel Fellowship, USNC-URSI (2024)
Lloyd W. Crawford Fellowship, UT Space Institute (UTSI) (2022 - 2023)
Best Paper Award, 1st International Conference on Artificial Intelligence and Data Analytics (CAIDA), Riyadh, Saudi Arabia (2021)
Deanship Scholarship (M.S.), King Fahd University of Petroleum and Minerals (KFUPM) (2019)
Technical Board Scholarship, Rajshahi University of Engineering & Technology (RUET) (2011 - 2014)

SPECIAL PROJECTS

1. **Kinetic and Reactor Modeling for Naphtha Catalytic Reforming**
   Sponsor: City and Regional Planning Department, KFUPM
   Role: Graduate Student (May 2020 - Apr 2022)
   Contribution: Literature survey, numerical simulation of differential equations, and parameter estimation (Completed)

REFERENCES

1. Professor Reza Abedi, UTK
2. Professor Trevor Moeller, UTSI
3. Professor Di Zhou, UTK